

KARTIK JOSHI

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PROFILE

AI Software Engineer at Algoleap with hands-on experience building production-grade LLM pipelines, RAG systems, and automated document intelligence solutions. Proficient in Python, LangChain, LangGraph, OpenAI models, FastAPI, and vector databases. Equally capable in Data Science workflows — from EDA and feature engineering to model deployment. Proven track record delivering client-facing AI products across real estate, CRM, and computer vision domains.

TECHNICAL SKILLS

AI / LLM: LLMs, RAG, Agentic Workflows, Prompt Engineering, LangChain, LangGraph, Vector Embeddings, Semantic Search, Computer Vision

Data Science: EDA, Feature Engineering, Scikit-Learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, Power BI

Backend & APIs: Python, FastAPI, Flask, REST APIs, JavaScript, SQL

Cloud & DevOps: AWS, Docker, Git

Programming: Python, C++, Java

Databases: PostgreSQL, MongoDB, FAISS, ChromaDB

WORK EXPERIENCE

AI Software Engineer · Algoleap · Hyderabad

Aug 2025 – Present

Client Project: CBRE – AI Inspection & Valuation App

- Engineered an end-to-end automated property report generation system ingesting structured form data, raw images, and multi-format documents — reducing manual report preparation time by ~70%.
- Designed vector storage and retrieval architecture using FAISS + LangChain + OpenAI embeddings, enabling high-speed similarity search across 500+ property documents.
- Built a GPT-4o Vision engine to auto-detect, classify, and describe visual elements from property photos, significantly cutting manual annotation effort.
- Developed high-performance FastAPI endpoints for AI model integration, handling concurrent requests with optimized response pipelines.
- Applied advanced prompt engineering and preprocessing to minimize hallucinations and improve extraction accuracy across Land, Residential, and Commercial property types.

General Engineering

- Designed and deployed scalable AI automation pipelines using OpenAI models and LangChain; collaborated with senior engineers to ensure smooth production rollout.
- Conducted data preprocessing, model evaluation, and prompt optimization to improve output quality and reduce latency across multiple client workstreams.

PROJECTS

AI Blueprint Analyzer — *FastAPI · LangChain · GPT-4o Vision · Streamlit*

- Built an AI system to analyze architectural blueprints, auto-extracting rooms, dimensions, and layout details using GPT-4o Vision combined with LLMs.
- Implemented conversational Q&A with voice input for natural language interaction over floor plans; CBRE-inspired split-screen UI for professional client use.

Salesforce MCP Assistant — *Model Context Protocol · FastAPI · LangChain · Streamlit*

- Built an intelligent assistant using Model Context Protocol (MCP) to connect LLMs with live Salesforce CRM data for natural language-driven query, create, and update operations.
- Integrated FastAPI backend with Salesforce SDK and LangChain; Streamlit frontend with real-time data visualization, built for an enterprise client environment.

GitHub Repo Analyzer — *LangChain · OpenAI · FastAPI · Streamlit · Mermaid*

- Developed an AI-driven repo analyzer (built for Vialto Partners) that auto-clones codebases and enables context-aware Q&A using RAG over source files.

- Auto-generated Mermaid architecture and flowchart diagrams from repo structure; natural language code Q&A via Streamlit UI.

Cold Email Generator — *LangChain · LLaMA 3.1 · ChromaDB · Groq · Streamlit*

- Built a personalized outreach tool that ingests a company careers URL, extracts job listings, matches them with portfolio links via ChromaDB embeddings, and drafts tailored emails using LLaMA 3.1 on Groq.
- Modular pipeline: URL loader → text extractor → embedding index → email generation UI, designed for easy future extension.

E D U C A T I O N

Bachelor of Computer Applications (BCA) · Graphic Era University, Haldwani · **CGPA 9.35 / 10**

Aug 2022 – Apr 2025

C E R T I F I C A T I O N S

- Introduction to Generative AI – Google Cloud / Simplilearn
- Chatbot Using AI – Coursera
- Data Analytics Basics for Everyone & Introduction to SQL – IBM / edX
- Data Analytics & Visualization Job Simulation – Accenture / Forage